

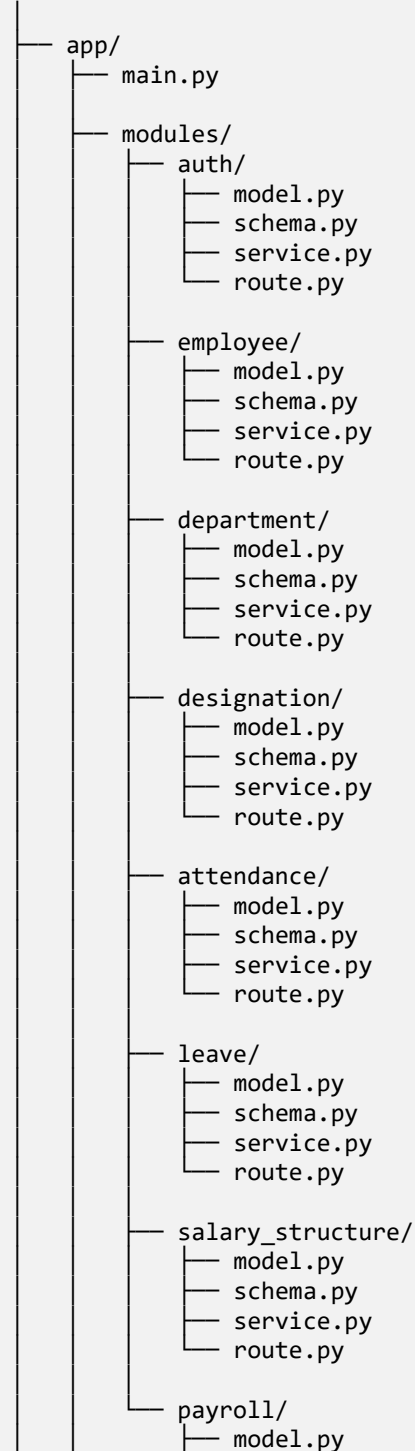
Employee Management & Payroll System

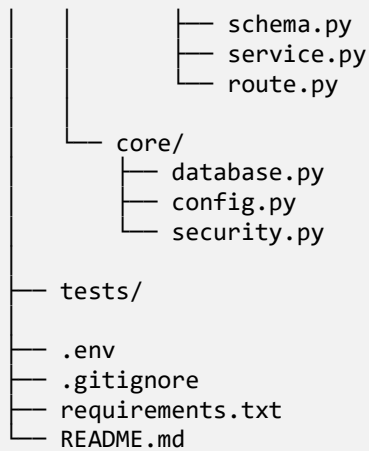
Final Module-Based Project Structure

1. Project Structure

The final structure organizes the backend into self-contained feature modules, each following the same internal layout, plus a shared core/ for cross-cutting concerns.

Employee-Management-Payroll-System/





2. What Each Part Does

modules/

Contains the actual business modules of the system. Each module is self-contained and holds everything related to that piece of functionality:

- `employee/`
- `attendance/`
- `leave/`
- `payroll/`

(and similarly: `auth/`, `department/`, `designation/`, `salary_structure/`)

model.py

Defines the database model/table using SQLAlchemy.

```
Employee model → Employee table
Payroll model → Payroll table
```

Example:

```
class Employee(Base):
    __tablename__ = "employees"
    ...

class Payroll(Base):
    __tablename__ = "payroll"
    ...
```

schema.py

Defines the data expected from or returned by the API, using Pydantic.

```
EmployeeCreate
EmployeeResponse
```

Example:

```
class EmployeeCreate(BaseModel):
    ...

class EmployeeResponse(BaseModel):
    ...
```

service.py

Contains the actual functionality / business logic for the module.

Example:

```
class EmployeeService:
    def create_employee(...): ...
    def get_employee(...): ...
    def update_employee(...): ...

class PayrollService:
    def calculate_salary(...): ...
    def generate_payroll(...): ...
```

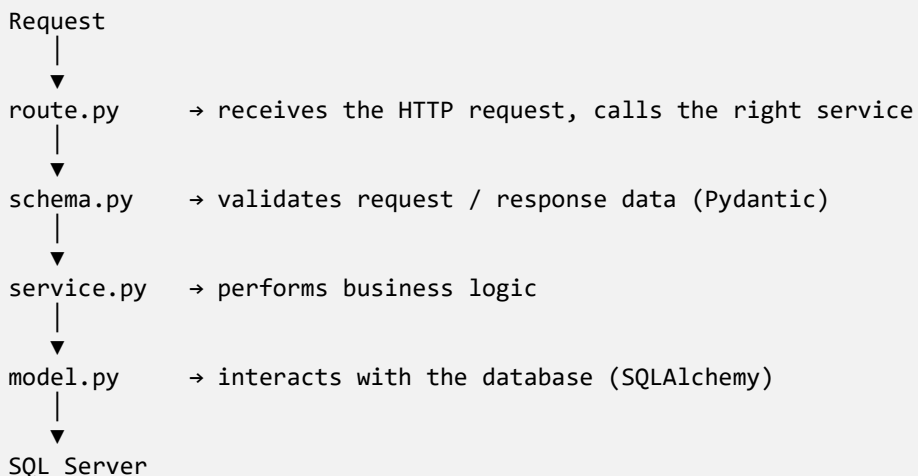
route.py

Contains the API endpoints, and calls the appropriate service.

```
POST /employees
GET /employees/{id}
PUT /employees/{id}
```

Request Flow

Every request moves through the same pipeline, regardless of module:



core/

Contains resources shared by all modules:

database.py — SQL Server connection / session management

config.py — application configuration

security.py — authentication and security utilities

main.py

Creates the FastAPI application and registers all module routes.

tests/

Contains tests for the APIs and business logic.

This layout keeps each feature (employee, attendance, leave, payroll, etc.) independently testable and easy to extend — adding a new module means adding a new folder with the same four files, without touching existing modules.